Executive Performance Summary of Plastic Recycling Plant

During the five-day operating window (September 01–05, 2026), the plant converted **810.26 metric tonnes** of incoming post-consumer polymer waste into circular manufacturing feedstock.

- **Material Recovery:** The wash plant achieved an **85.75% recovery yield**, producing 712.88K kg of clean washed flake against 96.57K kg of reject waste.
- **Circular Formulation:** True Post-Consumer Recycled (PCR) content in compounded pellets averaged **86.10%**, well above European regulatory thresholds for recycled packaging.
- **Decarbonization Value:** Plant output displaced virgin fossil polymers, delivering **355.89 metric tonnes of avoided Scope 3** CO_2e for downstream packaging customers.
- **Commercial Consistency:** Finished product pricing remained steady across sectors, yielding an average realized price between **€1,887.91/t** and **€1,940.86/t**.

Operational Diagnostics & Root Cause Analysis

- `Intake (810.26 t) ──> Wash Lines (85.75% Yield) ──> Extrusion (0.81M kg) ──> QA (57.6% Prime`

1. Wash Line Capacity Imbalance

Line 2 processed **442.56K kg** of clean flake (61.67K kg reject), while Line 1 yielded **270.32K kg** (34.90K kg reject). Line 2 is currently shouldering **62% of the plant's wash volume**. Both lines maintained equivalent recovery yields (~87.8% and ~88.5%), indicating consistent wet friction washing and hydro cyclone separation efficiency despite the volume disparity.

2. Compounding Quality Disruption

Extrusion throughput scaled from **0.10M kg** on Day 1 to peak runs of **0.23M kg/day** on Days 3 and 4. Extruder Specific Energy Consumption (SEC) peaked at **473 kWh/t** on Day 2 during initial barrel heating and purge, stabilizing down to **430 kWh/t** as throughput settled.

However, Quality Assurance lot testing on the 92 compounding batches reveals a critical bottleneck:

- **57.61% (53 lots)** met Prime Pass specifications.
- **32.61% (30 lots)** were **Downgraded** due to melt flow index (MFI) drift or tensile variation.
- **9.78% (9 lots)** suffered complete out-of-spec rejection.

3. Direct Impact on Bag Conversion Reliability

The downstream conversion facility logged **5,345 total minutes of machine downtime** and a **6.45% scrap rate**. Two root causes account for **75.8% of all line stoppages**:

- **Seal Failure:** 2,435 minutes
- **Bubble Burst:** 1,616 minutes

The high proportion of downgraded polymer lots (32.61%) directly correlates with blown-film conversion instability. Unstabilized PCR molecular weight distributions create melt-fracture and uneven film gauge, triggering repeated bubble collapse and poor heat-seal integrity on automated bag conversion machinery.